Packet Tracer – Simulation Mode: ARP, ICMP, and Packet Flow

Objectives

Part 1: Build a Simple Network Topology

Part 2: Configure Static IP Addressing

Part 3: Use Simulation Mode

Part 4: Observe ARP and ICMP Packet Flow

Background / Scenario

One of the most powerful features of Cisco Packet Tracer is **Simulation Mode**. Unlike Real-Time Mode, Simulation Mode allows you to slow down network communication and observe exactly how packets travel between devices.

In this lab, you will connect two PCs through a switch, configure static IP addresses, and use Simulation Mode to watch the exchange of **ARP (Address Resolution Protocol)** and **ICMP (Internet Control Message Protocol)** packets generated by the ping command.

This lab focuses on:

- Understanding how ARP resolves MAC addresses.
- Observing ICMP Echo Requests and Echo Replies.
- Following packet movement through a switch.
- Learning the sequence of events during network communication.

You are building a deeper understanding of how data moves across a network, an essential skill for future CCNA routing and switching labs.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. From **End Devices**, drag two **PCs** onto the workspace.

c. From **Network Devices → Switches**, drag one **2960 Switch** onto the workspace.

d. Rename the devices:

- PC0 → **PC-A**
- PC1 → **PC-B**
- Switch0 → **S1**

Step 2: Connect the Devices

Since PCs are connecting to a switch, you will use **Copper Straight-Through** cables.

a. Click **Connections** (lightning bolt icon).

b. Select **Copper Straight-Through**.

c. Connect:

- PC-A → FastEthernet0
- S1 → FastEthernet0/1

d. Connect:

- PC-B → FastEthernet0
- S1 → FastEthernet0/2

Wait until all link lights turn **green**.

Your physical topology is now complete.

Part 2: Configure Static IP Addressing

Since there is no DHCP server, you must manually assign IP addresses.

Step 1: Configure PC-A

a. Click **PC-A**.

b. Go to the **Desktop** tab.

c. Click **IP Configuration**.

d. Select **Static**.

Enter:

- IP Address: **192.168.1.1**
- Subnet Mask: **255.255.255.0**
- Default Gateway: *(leave blank)*

Close the window.

Step 2: Configure PC-B

a. Click **PC-B**.

b. Go to the **Desktop** tab.

c. Click **IP Configuration**.

d. Select **Static**.

Enter:

- IP Address: **192.168.1.2**
- Subnet Mask: **255.255.255.0**
- Default Gateway: *(leave blank)*

Close the window.

Why No Default Gateway?

A default gateway is only required when communicating with devices outside the local network.

Since both PCs belong to the **192.168.1.0/24** subnet, they can communicate directly through the switch.

Part 3: Use Simulation Mode

Step 1: Switch to Simulation Mode

a. Locate the mode selector in the lower-right corner of Packet Tracer.

b. Click **Simulation**.

The Event List window should appear.

Step 2: Clear Existing Events

a. Click **Delete** or **Reset Simulation** (depending on your Packet Tracer version).

b. Ensure the Event List is empty before continuing.

This allows you to observe only the packets generated during this lab.

Part 4: Observe ARP and ICMP Packet Flow

Step 1: Generate a Ping

a. Click **PC-A**.

b. Go to the **Desktop** tab.

c. Open **Command Prompt**.

d. Type:

ping 192.168.1.2

Press **Enter**.

Notice that the Event List now contains several packets.

Step 2: Capture the ARP Process

Click **Capture/Forward** several times.

You should observe:

1. PC-A sends an **ARP Request**.
2. The switch floods the ARP Request to all ports.
3. PC-B receives the request.
4. PC-B sends an **ARP Reply** back to PC-A.
5. The switch forwards the reply to PC-A.

The ARP process allows PC-A to learn PC-B's MAC address.

Step 3: Observe ICMP Communication

Continue clicking **Capture/Forward**.

After ARP completes, you should see:

1. ICMP Echo Request leaves PC-A.
2. The switch forwards the packet to PC-B.
3. PC-B receives the packet.
4. PC-B sends an ICMP Echo Reply.
5. The switch forwards the reply back to PC-A.

The ping operation is now complete.

What Is ARP?

Address Resolution Protocol (ARP) translates an IP address into a physical MAC address.

Example:

IP Address	MAC Address
192.168.1.2	00D0.BC12.3456

Without ARP, devices would know the destination IP address but would not know where to send the Ethernet frame.

What Is ICMP?

Internet Control Message Protocol (ICMP) is used for network testing and diagnostics.

The ping command uses:

- ICMP Echo Request
- ICMP Echo Reply

If the destination responds, communication is working correctly.

Typical Packet Flow

```
PC-A
  |
  |  ARP Request
  V
 Switch
  |
  |-------------------->
  |                PC-B
  |<--------------------
  |  ARP Reply
  |
  |  ICMP Echo Request
  V
 Switch
  |
  |-------------------->
  |                PC-B
  |<--------------------
  |  ICMP Echo Reply
  |
PC-A
```

Why Does ARP Appear Before ICMP?

The first ping usually takes slightly longer because the sender must first discover the destination MAC address.

The sequence is:

1. ARP Request
2. ARP Reply
3. ICMP Echo Request
4. ICMP Echo Reply

Future pings may skip the ARP process because the MAC address is temporarily stored in the ARP cache.

Troubleshooting Tips

If no packets appear:

✓ Make sure you switched from **Real-Time** to **Simulation** Mode.

✓ Verify the Event List was cleared before testing.

✓ Confirm both PCs have valid IP addresses.

✓ Check that the link lights are green.

✓ Ensure both devices are in the same subnet.

✓ Use **Capture/Forward** to step through each packet individually.

What You Learned

- How to use Packet Tracer's Simulation Mode.
- How ARP resolves IP addresses into MAC addresses.
- How ICMP is used by the ping command.
- The sequence of ARP and ICMP communication.
- How a switch forwards packets across a LAN.
- Why the first ping may take slightly longer than subsequent pings.

Challenge Extensions (Optional)

Try modifying the lab:

1. Perform a second ping after the first completes.
    - Does ARP occur again?
    - Why or why not?
2. Add a third PC connected to the switch.

    - Observe how the ARP Request is broadcast.

3. Change PC-B's IP address to **192.168.2.2**.
    - What happens during the simulation?
4. Add a router between two different networks and observe how packets travel through multiple devices.
5. Use the **Auto Capture/Play** button instead of **Capture/Forward** and compare the results.